

Summary

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Definition

If a function f is **differentiable** at $x = a$, then a **linear approximation to f at a** is a function given by

$$L_a(x) = f(a) + f'(a)(x - a)$$

Note 1: When the value of a is understood, the subscript is sometimes dropped from the notation. In this case, it is written as just $L(x)$ instead of $L_a(x)$.

Note 2: The graph of L is the line tangent to the graph of f at the point where $x = a$.

Note 3: If x is near a , the value $f(x)$ can be approximated by the value of $L(x)$.

Note 4: If the graph of f is concave down *on the interval* between a and x , then the approximation $f(x) \approx L(x)$ is an overestimate. If the graph of f is concave up *on the interval* between a and x , then the approximation $f(x) \approx L(x)$ is an underestimate. (The concavity has to be consistent across the interval, not just at a point.)

A **differential**, df of f at x is given by

$$df = f'(x)dx$$

Note 3: If we consider a point x and if L is the linear approximation of f at x , then for any point $x + dx$ the following holds

$$f(x + dx) \approx L(x + dx) = f(x) + f'(x)dx = f(x) + df$$

Note 4: The **increment** of f , Δf , is given by

$$\begin{aligned} \Delta f &= f(x + dx) - f(x) \approx L(x + dx) - f(x) = \\ &= f(x) + f'(x)dx - f(x) = f'(x)dx = df \end{aligned}$$

Therefore, $\Delta f \approx df$.